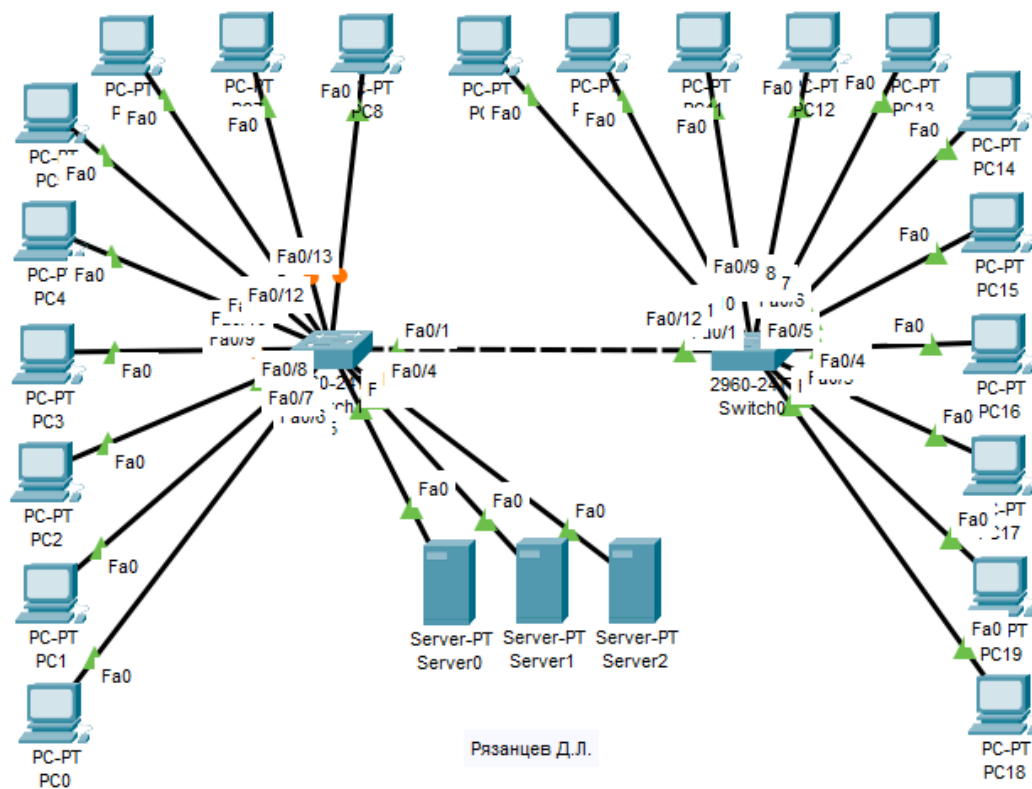
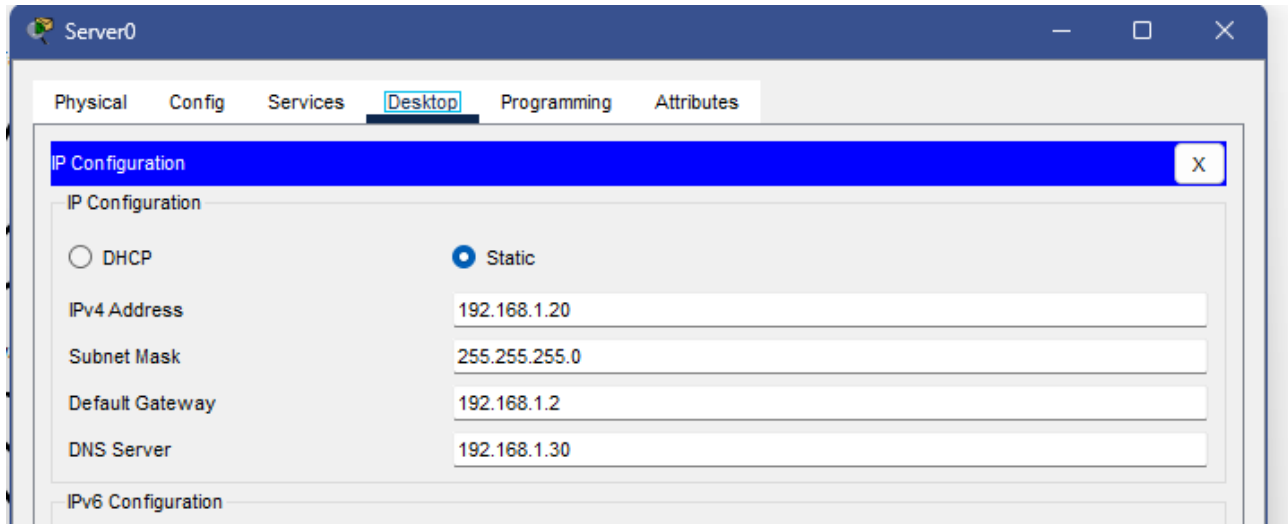


3. Соединяем устройства: коммутаторы - через *Copper Cross-Over*, остальные - через *Copper Straight-Through*



4. Настроим IP-конфигурацию для одного из серверов:



5. Настроим DHCP-сервер ( Services/DHCP ):

The screenshot shows the 'Services' tab for 'Server0'. On the left, a list of services includes HTTP, DHCP (selected), DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, and Radius EAP. The main area is titled 'DHCP' and contains the following configuration fields:

- Interface: FastEthernet0
- Service: ☒ On, ☐ Off
- Pool Name: serverPool
- Default Gateway: 192.168.2.1
- DNS Server: 192.168.1.30
- Start IP Address: 192, 168, 1, 0
- Subnet Mask: 255, 255, 255, 0
- Maximum Number of Users: 256
- TFTP Server: 0.0.0.0
- WLC Address: 0.0.0.0

Below the fields are 'Add', 'Save', and 'Remove' buttons. At the bottom, a table lists the configured DHCP pool:

| Pool Name  | Default Gateway | DNS Server  | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|------------|-----------------|-------------|------------------|-------------|----------|-------------|-------------|
| serverPool | 192.168....     | 192.168.... | 192.168....      | 255.255.... | 256      | 0.0.0.0     | 0.0.0.0     |

6. Убедимся в работоспособности DHCP-реквестов на PC0 :

The screenshot shows the 'IP Configuration' window for PC0. The 'DHCP' radio button is selected, and the 'Static' option is disabled. The configuration details are as follows:

- IPv4 Address: 192.168.1.1
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1
- DNS Server: 192.168.1.30

A message at the top right states 'DHCP request successful.'

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

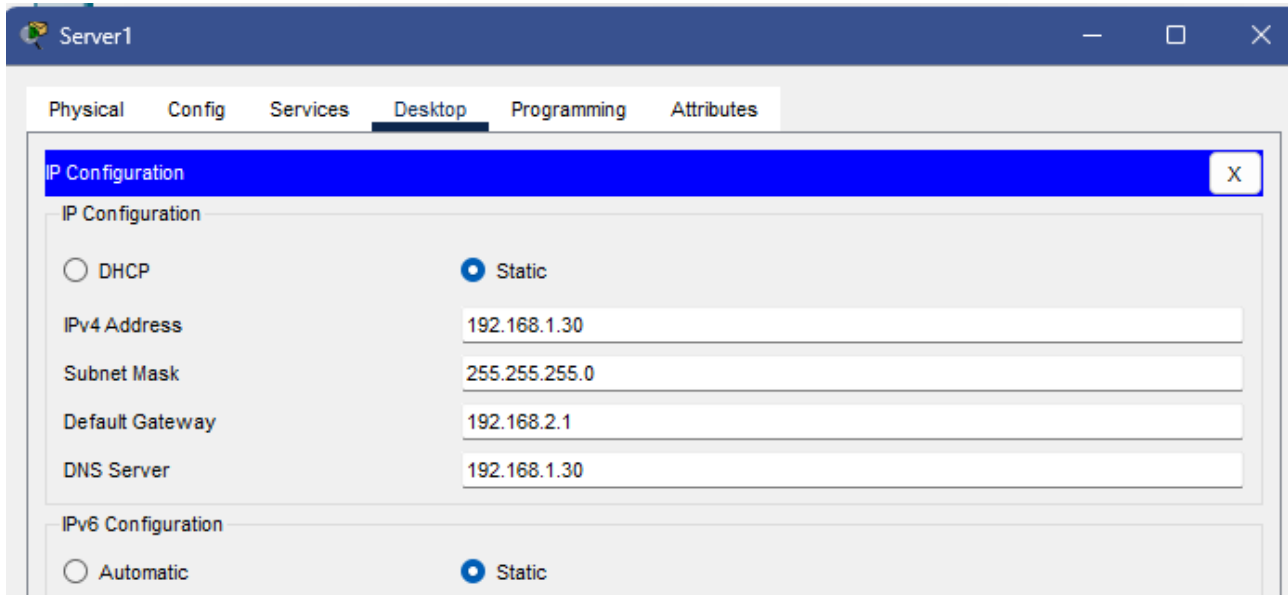
Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

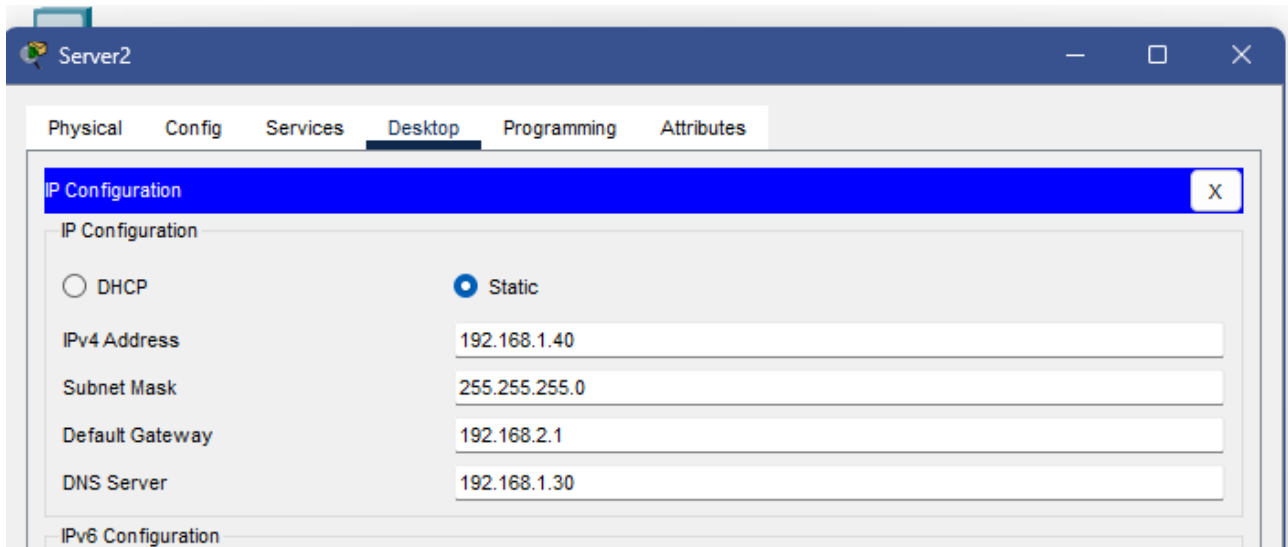
Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

7. Настроим IP-конфигурацию для второго сервера (DNS):



8. Настроим IP-конфигурацию для третьего сервера (DNS):



9. На первом сервере во вкладке *Services* -> *HTTP* изменим файл `index.html`:

```
<html>
  <center><font size='+2' color='blue'>Cisco Packet Tracer</font></cente
  <hr>Welcome to Cisco Packet Tracer. Opening doors to new opportunities
  <p>Quick Links:
  <br><a href='helloworld.html'>A small page</a>
  <br><a href='copyrights.html'>Copyrights</a>
  <br><a href='image.html'>Image page</a>
  <br><a href='cscoptlogo177x111.jpg'>Image</a>
  <br><b>Hello Polytech!</b>
</html>
```

10. Добавим на DNS A-Record для 192.168.1.20:

The screenshot shows the 'Server1' configuration window with the 'Services' tab selected. In the 'SERVICES' list on the left, 'DNS' is highlighted. The main panel shows the 'DNS' configuration. The 'DNS Service' is set to 'On'. Under 'Resource Records', the 'Name' field is empty, the 'Type' is set to 'A Record', and the 'Address' field is empty. Below these fields are 'Add', 'Save', and 'Remove' buttons. At the bottom, a table lists the existing records.

| No. | Name      | Type     | Detail       |
|-----|-----------|----------|--------------|
| 0   | www.hi.ru | A Record | 192.168.1.20 |

11. Убедимся в доступности всех трех серверов:

```
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time=6ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms

Control-C
^C
C:\>ping 192.168.1.30

Pinging 192.168.1.30 with 32 bytes of data:

Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.30:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
^C
C:\>ping 192.168.1.40

Pinging 192.168.1.40 with 32 bytes of data:

Reply from 192.168.1.40: bytes=32 time<1ms TTL=128
Reply from 192.168.1.40: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.40:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 2ms

Control-C
^C
C:\>
```

12. Убедимся, что HTML страница доступна по адресу `www.hi.ru`:

